

Node congestion and bisection width of a k -ary n -Cube

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Abstract

Edge congestion and node congestion which are defined using shortest path routing for all-to-all communication are in general difficult to find. In this paper, we not only calculate these measures in k -ary n -cubes but also aim at finding optimal routing algorithms such that the maximum edge congestion and maximum node congestion of a k -ary n -cube are respectively minimized. We show that k -ary n -cubes are uniformly edge-congested and uniformly node-congested. Furthermore, our proposed routing algorithms are shown to be optimal. By using edge congestion, we can provide a lower bound for bisection width of k -ary n -cubes for k odd.

Keywords: Bisection width, node congestion, edge congestion, forwarding index, shortest path routing, k -ary n -cubes.

1 Introduction

Since each vertex in an interconnection network corresponds to a processor or communication device, the congestion of a vertex in all-to-all communication can affect the network performance. Similarly, edges with high congestion are considered as bottlenecks and will delay communication time. A network with relatively balanced communication load of nodes and/or edges is preferred. The node congestion problem was first proposed by Chung et al. [2] and was called the *forwarding*

index problem. The edge congestion problem was studied in [4, 5]. In this paper, our main concern is to find shortest path routing algorithms which minimize the maximum node congestion and maximum edge congestion of a k -ary n cube.

To define node and edge congestion, we assume that messages exchanged between all pairs of vertices are equally probable. We thus consider routings between all pairs of vertices for calculating congestion and confine our discussion on **shortest path routing**. Throughout this paper, we use V and E to denote the vertex set and the edge set, respectively, of a network specified from the context without ambiguity. Let G denote a network. A routing algorithm π for the network G is defined as a function assigning each $(x, y) \in V \times V$ to only one path from x to y , denoted by $P_\pi(x, y)$, and sometimes written as (x, y) -path. We define the *node congestion* and *edge congestion* of $v \in V$ and $e \in E$ under a specified routing algorithm π , denoted by $c_\pi(v)$ and $c_\pi(e)$, respectively, as follows:

$$c_\pi(v) = |\{(x, y) \mid x, y \in V, v \text{ is in } P_\pi(x, y) \text{ and } v \neq x, y\}|,$$

$$c_\pi(e) = |\{(x, y) \mid x, y \in V, e \text{ is in } P_\pi(x, y)\}|.$$

Let $c_\pi(V)$ and $c_\pi(E)$ denote the node congestion and edge congestion of G under the routing algorithm π . We define

$$c_\pi(V) = \max\{c_\pi(v) \mid v \in V\},$$

$$c_\pi(E) = \max\{c_\pi(e) \mid e \in E\}.$$

Let $c_V(G)$ and $c_E(G)$ denote the node congestion and edge congestion of G , respectively, which are

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defined as follows:

$$c_V(G) = \min\{c_\pi(V) \mid \forall \text{ routing algorithm } \pi \text{ of } G\},$$

$$c_E(G) = \min\{c_\pi(E) \mid \forall \text{ routing algorithm } \pi \text{ of } G\}.$$

A shortest path routing algorithm that achieves $c_V(G)$ is called an *optimally node-congested routing algorithm*. Similarly, we define *optimally edge-congested routing algorithm*. If there is a routing algorithm π^* achieving $c_{\pi^*}(v) = c_V(G)$ for all $v \in V$, G is said to be *uniformly node-congested*. Furthermore, the algorithm π^* is an optimally node-congested routing algorithm. Similarly, we define *uniformly edge-congested* networks.

The *bisection width* of a graph G , denoted by $\omega(G)$, is the minimum number of edges to be removed to disconnect the graph into two (not necessarily connected) subgraphs with $\lceil |V|/2 \rceil$ and $\lfloor |V|/2 \rfloor$ vertices, respectively. The problem of finding the bisection width of a graph is NP-hard. The bisection width of an interconnection network is a critical factor in determining the speed with which the network can perform a computation and the area needed to layout this network.

Bisection width and edge congestion are related in the following way [4]:

$$c_E(G) \geq \begin{cases} \lceil \frac{|V|^2}{2\omega(G)} \rceil & \text{for } |V| \text{ even,} \\ \lceil \frac{|V|^2-1}{2\omega(G)} \rceil & \text{for } |V| \text{ odd.} \end{cases} \quad (1)$$

Though both bisection width and edge congestion are difficult to find, we sometimes can use (1) to find both of them for some graphs such as crossed cubes [1] or at least bounds for them.

2 Preliminaries

Given a routing algorithm π and an edge $e = (u, v)$, in order to calculate $c_\pi(e)$ we distinguish different orientation of e . We write $[u, v]$ and $[v, u]$ as traversing from u to v and from v to u , respectively. We use the following concept to calculate $c_\pi(e)$:

$$c_\pi(e) = |\{(x, y)\text{-pair} \mid P_\pi(x, y) \text{ routes through } [u, v] \text{ or } [v, u]\}|.$$

Given a path P , we use $|P|$ to denote the length of P . Each shortest path P adds $c_\pi(e)$ by 1 for all

$|P|$ edges e in P and adds $c_\pi(v)$ by 1 for all $|P| - 1$ nodes in P which are distinct from the source and the destination. It follows that the sum of the all-pair shortest distance, denoted by T , can be written as

$$T = \sum_{e \in E} c_\pi(e) = \sum_{v \in V} c_\pi(v) + |V| \cdot (|V| - 1). \quad (2)$$

Using (2), we can get lower bounds for c_V and $c_E(G)$ as follows:

$$c_V(G) \geq \lceil \frac{T - |V| \cdot (|V| - 1)}{|V|} \rceil$$

$$c_E(G) \geq \lceil \frac{T}{|E|} \rceil.$$

Let π be a shortest path routing algorithm of G . We establish a relationship between $c_\pi(e)$ and $c_\pi(v)$ in the following lemma.

Lemma 1 *Let π be a routing algorithm of G . Then,*

$$2c_\pi(v) + 2(|V| - 1) = \sum_{e \ni v} c_\pi(e) \quad \text{for all vertices } v,$$

where $e \ni v$ means for all edges incident at v .

Proof: For each vertex v , we distinguish the following three types of paths traversing v , that is, $|V| - 1$ paths originating from v , $|V| - 1$ paths ending at v and $c_\pi(v)$ paths passing through v as an internal node. Note that each of those $c_\pi(v)$ paths adds $c_\pi(e)$ of two edges e which are incident at v by 1. Therefore, we have $2c_\pi(v) + 2(|V| - 1) = \sum_{e \ni v} c_\pi(e)$. The lemma follows.

Using Lemma 1, we immediately obtain the following corollary.

Corollary 1 *Let G be a regular network of degree δ , that is, each vertex has degree δ . If G is uniformly edge-congested, then the following statements hold:*

- (i) G is uniformly node-congested;
- (ii) $c_V(G) = \delta/2 \cdot c_E(G) - |V| + 1$;
- (iii) if a routing algorithm π achieves uniform edge congestion of G , then π also achieve uniform node congestion.

The definition of Cartesian product of two graphs. The *Cartesian product* of two graphs

$G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, denoted by $G_1 \times G_2$, is the graph $G = (V, E)$, where

$$\begin{aligned} V &= \{xy \mid x \in V_1, y \in V_2\}, \\ E &= \{(x_1y, x_2y) \mid (x_1, x_2) \in E_1, y \in V_2\} \\ &\quad \cup \{(xy_1, xy_2) \mid x \in V_1, (y_1, y_2) \in E_2\}. \end{aligned}$$

The graph $G_1 \times G_2$ has $|V_1| \cdot |V_2|$ vertices and $|V_1| \cdot |E_2| + |E_1| \cdot |V_2|$ edges. We can similarly define Cartesian product of n graphs, and we write $\Pi_{i=1}^n G_i = G_1 \times G_2 \times \dots \times G_n$. In particular, $\Pi_{i=1}^n G$ is called the n -th Cartesian power of G .

Each node v of a k -ary n -cube Q_n^k is represented by an n -bit radix- k vector, where $k \geq 3$. Two nodes $u = u_{n-1}u_{n-2} \dots u_0$ and $v = v_{n-1}v_{n-2} \dots v_0$ are connected by an edge in Q_n^k if and only if $u_d = v_d \pm 1 \pmod{k}$ and $u_j = v_j$ for all $j \neq d$, where $0 \leq d \leq n-1$. Furthermore, we call (u, v) a $\text{dim-}d$ edge, and v the d -neighbor of u and vice versa. For example, let $u = 1012$ be a vertex in Q_4^3 . Then the 0-neighbors of u are given by 1010 and 1011. The k -ary n -cube Q_n^k is the n -th Cartesian power of C_k , denoted by $\Pi_{i=1}^n C_k$, where C_k denotes a cycle of length k .

In this paper, we show that Q_n^k is uniformly edge-congested. Although we can apply Corollary 1 to find the node congestion of these networks, we still include some proofs in this paper and think that the proof technique can be useful to be applied to networks which are not uniformly edge-congested.

3 Results

Note that $k \geq 3$ is assumed in this section. Edge congestion of k -ary n -cubes was solved in [4], as stated below.

Theorem 1 Q_n^k is uniformly edge-congested, and $c_E(Q_n^k) = k^{n-1} \lfloor k^2/4 \rfloor$.

Since $Q_n^k = \Pi_{i=1}^n C_k$, in order to find $c_V(Q_n^k)$ we first find $c_V(C_k)$. A routing algorithm for C_k is given as follows:

Algorithm τ for C_k .

Input: vertices x and y , where $0 \leq x, y \leq k-1$.

Output: a shortest path from x to y .

Method: Let $P_1 = \langle x, x \oplus 1, x \oplus 2, \dots, y \oplus 1, y \rangle$ and $P_2 = \langle x, x \ominus 1, x \ominus 2, \dots, y \oplus 1, y \rangle$, where $\oplus$ and $\ominus$ represent addition and subtraction with modulo k , respectively.

1. If k is even and $|x - y| = k/2$, route by P_1 .
2. If $1 \leq y - x \leq \lfloor k/2 \rfloor$ or $x - y \geq \lfloor k/2 \rfloor + 1$, route by P_1 .
3. If $y - x \geq \lfloor k/2 \rfloor + 1$ or $1 \leq x - y \leq \lfloor k/2 \rfloor$, route by P_2 .

Lemma 2 C_k is uniformly node-congested, and $c_V(C_k) = \lfloor (k-2)^2/4 \rfloor$.

Proof: Consider using the routing algorithm τ in C_k . First, consider k is odd. Note that $P_\tau(x, y)$ and $P_\tau(y, x)$ use the same edges with traversing in different orientation. It follows that for any vertex v , the number of paths passing through v as an internal node with a specified length $l \geq 2$ is $2(l-1)$. Since the diameter of C_k is $\lfloor k/2 \rfloor$, we have

$$c_\tau(v) = \sum_{l=2}^{\lfloor k/2 \rfloor} 2(l-1) = \lfloor k/2 \rfloor \cdot (\lfloor k/2 \rfloor - 1) = \lfloor (k-2)^2/4 \rfloor.$$

Next consider k is even. Note that $P_\tau(x, y)$ and $P_\tau(y, x)$ with $|P_\tau(x, y)| = k/2$ traverse different internal nodes. It follows that

$$c_\tau(v) = k/2 - 1 + \sum_{l=2}^{k/2-1} 2 \cdot (l-1) = (k-2)^2/4 = (k-2)^2/4.$$

Hence the lemma is proved.

Algorithm π for Q_n^k :

Input: vertices x and y .

Output: a shortest path from x to y .

Method: Change the rightmost differing bit iteratively by using the algorithm τ ,

i.e., for $i = 0$ to $n-1$, changing x_i to y_i by using the routing algorithm τ .

Theorem 2 Q_n^k is uniformly node-congested, and

$$c_V(Q_n^k) = \begin{cases} n/4 \cdot (k^{n+1} - k^{n-1}) - k^n + 1 & \text{for } k \text{ odd,} \\ n/4 \cdot k^{n+1} - k^n + 1 & \text{for } k \text{ even.} \end{cases}$$

Proof: Consider using the routing algorithm π in Q_n^k . Let $v = v_{n-1}v_{n-2} \dots v_0$ be any vertex in Q_n^k . Let (x, y) -pair route through v under π , where x and y are distinct from v . It follows that we can write $x = (x^3, x^2, x^1)$ and $y = (y^3, y^2, y^1)$, where

$$\begin{aligned} x^3 &= v_{n-1}v_{n-2} \dots v_{j+1}, \\ x^2 &= v_j v_{j-1} \dots v_{i+1} \hat{v}_i, \\ x^1 &= x_{i-1} x_{i-2} \dots x_0, \\ y^3 &= y_{n-1} y_{n-2} \dots y_{j+1}, \\ y^2 &= \hat{v}_j v_{j-1} \dots v_{i+1} v_i, \\ y^1 &= v_{i-1} v_{i-2} \dots v_0, \end{aligned}$$

where $\hat{v}_i \neq v_i$, $\hat{v}_j \neq v_j$, and $0 \leq i \leq j \leq n-1$.

We distinguish two cases of j . First, consider $i = j$. It follows that

$$\begin{aligned} & | \{ (x, y)\text{-pair} \mid (x, y)\text{-path routes through } v \\ & \quad \text{in a cycle } C_k \} | \\ &= n \cdot k^{n-1} \cdot c_V(C_k) = n \cdot k^{n-1} \cdot \lfloor (k-2)^2/4 \rfloor. \end{aligned} \quad (3)$$

Second, consider $j > i$. It follows that

$$\begin{aligned} & | \{ (x, y)\text{-pair} \mid (x, y)\text{-path routes through } v \\ & \quad \text{with } j > i \} | \\ &= \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} k^i \cdot k^{n-1-j} \cdot (k-1)^2 \\ &= (n-1) \cdot k^n - n \cdot k^{n-1} + 1, \end{aligned} \quad (4)$$

by using the formula $\sum_{i=0}^m k^i = (k^{m+1}-1)/(k-1)$. Therefore,

$$\begin{aligned} c_\pi(v) &= (3) + (4) \\ &= \begin{cases} n/4 \cdot (k^{n+1} - k^{n-1}) - k^n + 1 & \text{for } k \text{ odd,} \\ n/4 \cdot k^{n+1} - k^n + 1 & \text{for } k \text{ even.} \end{cases} \end{aligned}$$

which is independent of v . Hence the theorem follows.

Corollary 2 *The proposed routing algorithm π is an optimally node-congested routing algorithm.*

We can easily prove $\omega(Q_n^k)$ for k even by using (1) even though we can use similar proof technique for proving $\omega(Q_n)$ stated in [6].

Theorem 3 $\omega(Q_n^k) = 2k^{n-1}$ for k even.

Proof: It follows from Theorem 3 and (1) that $\omega(Q_n^k) \geq 2k^{n-1}$. On the other hand, we want to find a bisection cut of $2 \cdot k^{n-1}$ edges. It is observed that Q_n^k can be treated as $Q_{n-1}^k \times C_k$ and that C_k can be bisected by removing two edges $e_1 = (0, 1)$ and $e_2 = (k/2, k/2 + 1)$. Thus, removing edges of $Q_{n-1}^k \times e_1 \cup Q_{n-1}^k \times e_2$ bisects Q_n^k into two disjoint subgraphs of equal size, i.e., $Q_{n-1}^k \times P_{k/2}$, where $P_{k/2}$ is a path of $k/2$ vertices. Therefore, $\omega(Q_n^k) \leq 2k^{n-1}$ and the theorem follows.

It is still open for $\omega(Q_n^k)$ with k odd. We here provide a lower bound and an upper bound for $\omega(Q_n^k)$ with k odd.

Lemma 3 $\omega(Q_n^k) > \frac{2(k^n-1)}{k-1} \cdot \frac{k}{k+1}$ for k odd.

Proof: It follows from (1) and that

$$\begin{aligned} & k^{n-1}(k^2 - 1)/4 \\ & \geq \lceil (k^{2n} - 1)/[2\omega(Q_n^k)] \rceil \\ & > (k^{2n} - 1)/[2\omega(Q_n^k)]. \end{aligned}$$

Therefore, we have

$$\begin{aligned} & \omega(Q_n^k) \\ & > 2 \frac{k^{2n} - 1}{k^{n-1}(k^2 - 1)} = 2 \frac{k^n - 1}{k - 1} \frac{k + \frac{1}{k^{n-1}}}{k + 1} \\ & > 2 \frac{k^n - 1}{k - 1} \frac{k}{k + 1}. \end{aligned}$$

The lemma follows.

We use $Q_{n-1}^k(j)$, $0 \leq j \leq k-1$, to denote the subgraph of Q_n^k induced by the set of vertices v with $v_{n-1} = j$, which is a k -ary $(n-1)$ -cube. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$. If $V_1 \subseteq V$ and $E_1 \subseteq E$, we say $G_1 \subseteq G$. We say $G = G_1 \cup G_2$ if $G = (V_1 \cup V_2, E_1 \cup E_2)$.

Lemma 4 $\omega(Q_n^k) \leq \frac{2(k^n-1)}{k-1}$ for k odd.

Proof: We prove this lemma by induction on n . Since $Q_1^k = C_k$, its bisection width is 2. Assume that the lemma is true for Q_i^k with $i \leq n-1$ and $n \geq 2$. Let Q_i^k can be bisected by w_i edges, and $w_1 = 2$.

We will show that the lemma holds for Q_n^k . Consider a bisection of Q_n^k as follows. First, we remove the number of $k^{n-1} \dim(n-1)$ edges where each edge has one end in $Q_{n-1}^k(0)$ and the other end in $Q_{n-1}^k(k-1)$. Then Q_n^k is reduced to $Q_{n-1}^k \times P_k$, where P_k is a path of k vertices. Let G'_1 and G'_2 denote two subgraphs of Q_n^k given by $G'_1 = \cup_{i=0}^{\lfloor k/2 \rfloor - 1} Q_{n-1}^k(i)$ and $G'_2 = \cup_{j=\lfloor k/2 \rfloor}^{k-1} Q_{n-1}^k(j)$. It follows that a path from $u \in V(G'_1)$ to $v \in V(G'_2)$ must go through a vertex $x \in V(Q_{n-1}^k(\lfloor k/2 \rfloor))$.

Next, we find a cut to bisect $Q_{n-1}^k(\lfloor k/2 \rfloor)$. By the induction hypothesis, $Q_{n-1}^k(\lfloor k/2 \rfloor)$ can be bisected by at most w_{n-1} edges into two disjoint subgraphs G_1 and G_2 with $\lfloor k^{n-1}/2 \rfloor$ vertices and $\lceil k^{n-1}/2 \rceil$ vertices, respectively. Let $G_1^* = G'_1 \cup G_2$ and $G_2^* = G'_2 \cup G_1$. Note that $|V(G_1^*)| - |V(G_2^*)| = 1$. Furthermore, G_1^* and G_2^* are connected only by $\dim(n-1)$ edges between $Q_{n-1}^k(\lfloor k/2 \rfloor - 1)$ and $Q_{n-1}^k(\lfloor k/2 \rfloor)$, and between $Q_{n-1}^k(\lfloor k/2 \rfloor)$ and $Q_{n-1}^k(\lceil k/2 \rceil)$.

Finally, remove the number of $\lfloor k^{n-1}/2 \rfloor \dim(n-1)$ edges with one end of each edge in G_1 and the other in the corresponding $Q_{n-1}^k(\lfloor k/2 \rfloor - 1) \subseteq G'_1$, and then remove the number of $\lceil k^{n-1}/2 \rceil \dim(n-1)$ edges with one end of each edge in G_2 and the other in the corresponding $Q_{n-1}^k(\lceil k/2 \rceil)$. It follows that after removing these edges from Q_n^k , G_1^* is no longer connected to G_2^* . Hence, Q_n^k is bisected into two disjoint subgraphs G_1^* and G_2^* .

Furthermore, we note that

$$\begin{aligned} w_n &= k^{n-1} + \lfloor k^{n-1} / 2 \rfloor + \lceil k^{n-1} / 2 \rceil + w_{n-1} \\ &= 2k^{n-1} + w_{n-1} \quad \text{for } n \geq 2, \end{aligned}$$

and $w_1 = 2$. Solving this recurrence equation, we obtain $w_n = 2(k^n - 1) / (k - 1)$. Since $\omega(Q_n^k) \leq w_n$, the lemma follows.

We conclude Lemmas 3 and 4 in the following theorem.

Theorem 4 $\frac{2(k^n - 1)}{k - 1} \cdot \frac{k}{k + 1} < \omega(Q_n^k) \leq \frac{2(k^n - 1)}{k - 1}$ for k odd.

We list the above lower bound and upper bound for $k = 3, 5$ and various n in the following table:

n	3	4	5	6	7	8	9	10
$k = 3$	20	60	182	546	1640	4920	14762	44286
	26	80	242	728	2186	6560	19682	59048
$k = 5$	52	260	1302	6510	32552	162760	813802	4069010
	62	312	1562	7812	39062	195312	976562	4882812

The first and the second numbers in each cell of the table represent the lower bound and the upper bound, respectively.

4 Concluding Remarks

The k -ary n -cubes are shown to be node-symmetric and edge-symmetric. Furthermore, they are shown to be uniformly edge-congested and uniformly node-congested. A node-symmetric network is not necessarily uniformly edge-congested. For example, let G be a graph obtained by connecting two cycles of length 3, having the node sets $\{1, 2, 3\}$ and $\{a, b, c\}$, with three edges $(1, a)$, $(2, b)$ and $(3, c)$. Obviously, G is node-symmetric but not edge-symmetric. Moreover, G has a diameter 2 and $|E| = 9$. Since the sum of all-pair shortest distance is $42 (= 12 \times 2 + 18)$, which is not a multiple of $|E|$. Though G is node-symmetric, G is not uniformly edge-congested. On the other hand, an edge-symmetric network is not necessarily uniformly node-congested. Note that edge-symmetric networks are not necessarily regular. For example, $K_{1,n}$ is a tree of $n + 1$ vertices with one vertex connected to each of the remaining n vertices. $K_{1,n}$ is edge-symmetric and uniformly edge-congested, but neither regular nor uniformly node-congested. We wonder whether node-symmetric networks are uniformly

node-congested and whether edge-symmetric networks are uniformly edge-congested.

By applying Lemma 1 we cannot answer whether uniform node congestion implies uniform edge congestion for regular graphs. Can we add some properties to regular graphs such that the above statement holds? We conjecture that for edge-symmetric regular graphs, uniform node congestion implies uniform edge congestion. Since regular networks are in general preferred, it is an interesting problem to characterize regular graphs which are uniformly node-congested and/or uniformly edge-congested.

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